WEEK – 09

9. Packages – Roman to Integer

Basic Information

Experiment No: 09
Date: 18/10/2026
Title: Roman to Integer Conversion using Packages in Java

Objective / Aim

To design and implement a **Roman to Integer Conversion program** in Java using the concept of **Packages**. The program demonstrates the creation of a user-defined package, defining classes and methods, and converting Roman numeral strings into their corresponding integer values. It also demonstrates Java concepts such as switch statements, loops, conditional statements, string handling, and object creation.

System Requirements

Hardware
• Standard PC/Laptop
• Intel Core i3/i5 processor or equivalent
• Minimum 4 GB RAM
• Minimum 500 MB free storage

Software
• Operating System: Windows / Linux / macOS
• Java Development Kit (JDK 17 or JDK 21)
• Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
• Command Prompt / Terminal

Algorithm / Logic Description

1. Create a user-defined package named `helloworldpackage`.

2. Import the `Scanner` class to accept input from the user.

3. Create a class named `RomanConverter`.

4. Define a private method named `getValue()` to return the integer value of each Roman symbol.

5. Use a switch statement to assign values: I = 1, V = 5, X = 10, L = 50, C = 100, D = 500, and M = 1000.

6. Define a public method named `romanToInteger()` that accepts a Roman numeral string.

7. Initialize the result variable to zero and convert the input to uppercase.

8. Traverse each character using a for loop and get the current symbol value.

9. Compare the current symbol value with the next symbol value.

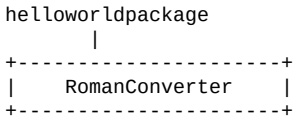
10. If the current value is smaller than the next value, subtract it from the result; otherwise, add it.

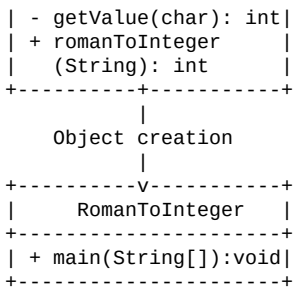
11. Return the final integer value.

12. Create the main class named `RomanToInteger`.

13. Create an object of `RomanConverter`, read a Roman numeral using Scanner, call `romanToInteger()`, and display the result.

UML Class Diagram





Relationship: The `RomanToInteger` class creates an object of the `RomanConverter` class to access the Roman numeral conversion method. Both classes belong to the same user-defined package.

Source Code

File Name: `RomanToInteger.java`

```
package helloworldpackage;

import java.util.Scanner;

class RomanConverter {

    // Method to get Roman symbol value
    private int getValue(char ch) {

        switch (ch) {
            case 'I': return 1;
            case 'V': return 5;
            case 'X': return 10;
            case 'L': return 50;
            case 'C': return 100;
            case 'D': return 500;
            case 'M': return 1000;
            default: return 0;
        }
    }

    // Method to convert Roman to Integer
    public int romanToInteger(String roman) {

        int result = 0;

        roman = roman.toUpperCase();

        for (int i = 0; i < roman.length(); i++) {

            int current = getValue(roman.charAt(i));

            if (i + 1 < roman.length()) {

                int next = getValue(roman.charAt(i + 1));

                if (current < next) {
                    result -= current;
                } else {
                    result += current;
                }

            } else {
                result += current;
            }
        }

        return result;
    }
}

// Main class
public class RomanToInteger {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        RomanConverter converter = new RomanConverter();
```

```
        System.out.println("===== ROMAN TO INTEGER CONVERSION =====");

        System.out.print("Enter a Roman Numeral: ");

        String roman = sc.nextLine();

        int result = converter.romanToInteger(roman);

        System.out.println("Roman Numeral: " + roman.toUpperCase());

        System.out.println("Integer Value: " + result);

        sc.close();
    }
}
```

Compilation & Execution Commands

Save the source code as `RomanToInteger.java` inside the `helloworldpackage` package.

```
javac -d . RomanToInteger.java
java helloworldpackage.RomanToInteger
```

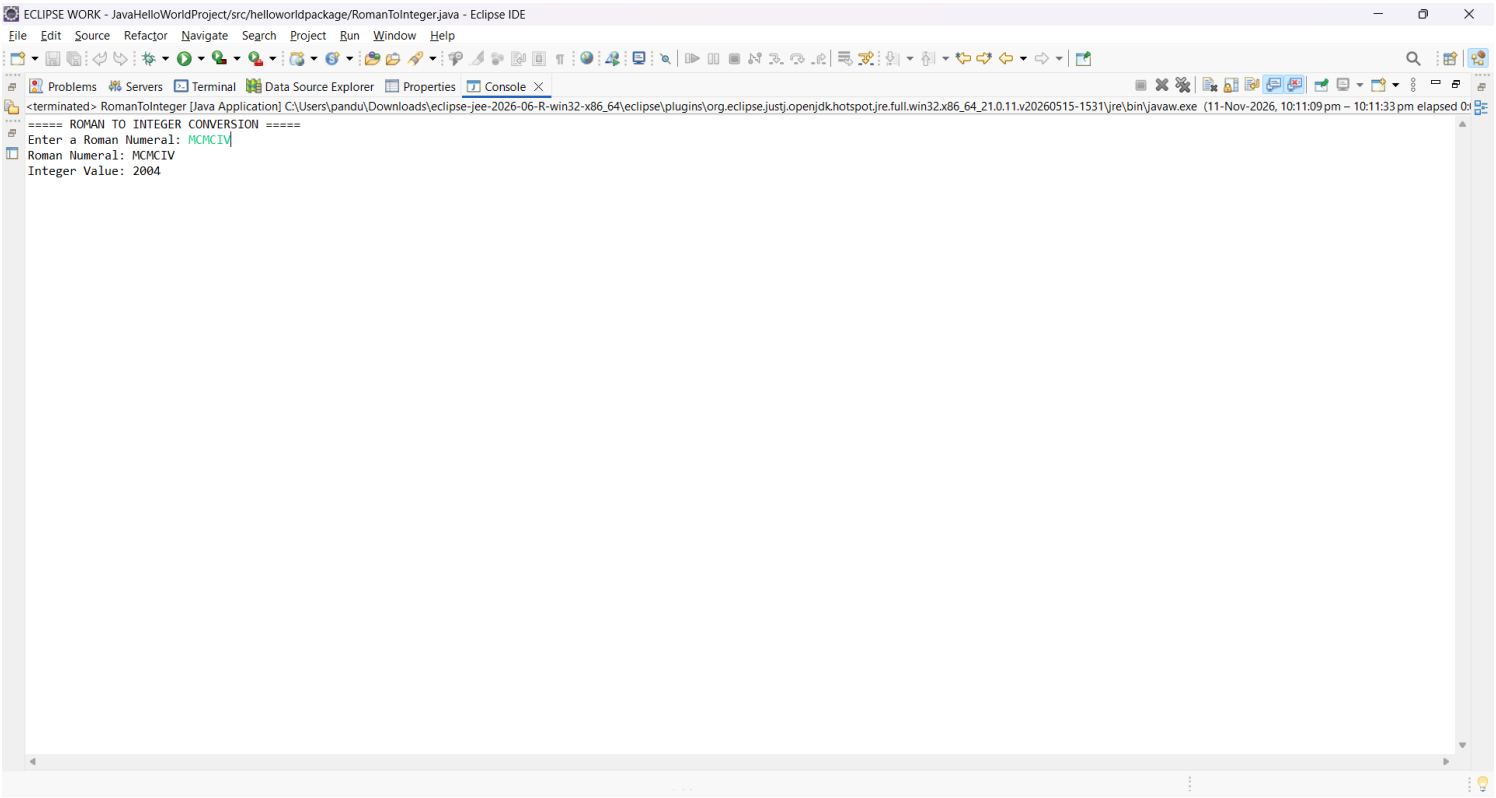
Output Screen

```
===== ROMAN TO INTEGER CONVERSION =====
Enter a Roman Numeral: MCMXCIV
Roman Numeral: MCMXCIV
Integer Value: 1994
```

Result

The **Roman to Integer Conversion program** was successfully implemented in Java using a user-defined package. The program demonstrates package declaration, class creation, object creation, methods, switch statements, loops, and string handling to convert Roman numerals into their corresponding integer values.

Note: This is a single-file implementation with both classes in the same package. It demonstrates package declaration and usage, but not importing a separate user-defined package. The conversion logic handles standard Roman numeral subtraction, but does not validate every Roman numeral formation rule.


ECLIPSE WORK - JavaHelloWorldProject/src/helloworldpackage/RomanToInteger.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Problems Servers Terminal Data Source Explorer Properties Console
<terminated> RomanToInteger [Java Application] C:\Users\pandu\Downloads\eclipse-jee-2026-06-R-win32-x86_64\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (11-Nov-2026, 10:11:09 pm – 10:11:33 pm elapsed 0:
===== ROMAN TO INTEGER CONVERSION =====
Enter a Roman Numeral: MCMCIV
Roman Numeral: MCMCIV
Integer Value: 2004